

Operations Research: theory and applications to networking

Birkhoff - Von Neumann Decomposition

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ICT4SS+CE

Sub-permutation Matrices

A $N \times N$ matrix $A = [a_{ij}]$ is said to be a sub-permutation matrix iff:

- $a_{ij} \in \{0, 1\}$
- $\sum_i a_{ij} \leq 1 \qquad \sum_j a_{ij} \leq 1$

Examples: $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

BvN decomposition

Given an $N \times N$ matrix $B = [b_{ij}]$, with non negative integer entries b_{ij} , decompose it as a sum of sub-permutation matrices:

$$B = \sum_{h=1}^K A_h$$

by minimizing the number K of employed sub-permutation matrices.

Relationship with matching?

Find the connection between the **Birkhoff–von Neumann decomposition** and the **matching problem**, specifically in the context of **cell scheduling in routers**.

How to generate instances

Generate 6x6 random matrices in which entries are i.i.d uniformly distributed over $\{0, 1, 2, \dots, 12\}$.